MIL-L-37704
30 September 1977

MILITARY SPECIFICATION

LIGHT, HEAD, DIAGNOSTIC, LEMPERT-STORZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adjustable headlight, of the Lempert-Storz design, suitable for use in operating and other medical procedures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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MILITARY

MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

No. 20719 LIGHT, HEAD

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 56A Standard For The Use of Inhalation Anesthetics.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 American Standard Dimensions of Plugs and Receptacles-General Purpose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, NY 10018.)

UNDERWRITERS' LABORATORIES, INC.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, LI, NY 11749; 207 East Ohio Street, Chicago, IL 60611; and 1655 Scott Boulevard, Santa Clara, CA 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Fire and casualty hazards. The electrical components, including transformer of the unit, specified herein shall comply with the fire and casualty hazards provisions of the applicable standards for operation in hazardous locations, Class 1, Group C, and as prescribed by the National Fire Protection Association NFPA No. 56A, and be an approved type for the purpose intended, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp, or label to the article, or issuance of a certificate. The stamp, label, seal or certificate of the Underwriters' Laboratories, Inc., or other testing laboratory approved by the contracting agency shall be considered acceptable evidence of such approval. However, in the addition to the aforementioned, the contracting agency reserves the right to require the contractor to submit a laboratory test report which shall include test results and methods used to determine conformance. Compliance with these requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

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3.2 Material.

3.2.1 Mirror. Mirror shall be fabricated from the highest quality optical glass, free from defects. Mirror shall be back surfaced, and suitable for the purpose intended.

3.2.2 Headband. Headband shall be fabricated from suitable durable, lightweight, non-staining nylon. Headband shall be lined with a washable cloth suspension lining.

3.2.3 Other material. All metal components shall be fabricated from austenitic or martensitic corrosion-resistant steel. All other material shall be of high quality, suitable in all respects for the intended purpose.

3.3 Style, design and dimensions. Shall be a headlight of the Lempert-Storz design, suitable for use in general medical and surgical procedures. The unit shall be furnished in a case, complete with headband, mirror, lamp, transformer, bayonet-type locking connectors, line cord with 3-prong plug, and an extra lamp. Adjustment of headband shall be easy and self-locking. The unit shall be designed to permit easy and effective adjustment by means of a spiral gear assembly that shall provide positive locked beam focusing. The focus of the headlight shall remain unchanged during normal movements of the operator, or when the mirror assembly is adjusted. Mirror assembly shall be adjustable by means of three supporting tension arms that shall permit both vertical and lateral adjustment. Headband shall permit rapid, easy adjustment without the use of locking or pressure nuts. Headband suspension lining shall effect weight distribution of the headlight, eliminating pressure points. Mirror shall be specially ground, with polished optical surface, full size (not slotted), and be readily removed and assembled to mirror metal housing. Mirror shall allow operator indirect or direct vision into surgical cavity in same plane as reflected light. The illumination of the headlight assembly shall be brillant and shadow free when connected to a nominal 115 volt, 60 Hz, electrical source.

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3.3.1 Transformer. Transformer shall be adapted to a primary of 115 volts, with a secondary output of 8 volts. Transformer shall be of the fixed voltage type and shall be wired with a 3-wire grounded electric system. Transformer shall be equipped with a flexible, insulated, 3-wire cord as shown on Defense Personnel Support Center (DPSC) Drawing No. 20719. A plug, in accordance with ANSI Standard C73.11, shall be attached to one end of the power cord. A large, red warning tag shall be securely attached to the cord, in proximity of the plug, which shall have prominent sized, boldfaced characters alerting the user not to operate the headlight with the plug provided, but to remove the plug and assemble an explosion proof plug to power cord, in accordance with NFPA No. 56A requirements, approved Underwriters' Laboratories (or other approved agency) for operation in class 1, group C locations when headlight is to be operated in any hazardous location. The cord carrying the secondary transformer voltage shall be permanently and securely attached to the transformer and shall be constructed of suitable insulated shielded cable exhibiting high tensile strength and good dielectric properties. Bayonet style lock connectors shall be provided for connecting and disconnecting the transformer assembly from the headlight proper. The bayonet connection shall not permit unintentional electrical and mechanical separation between the headlight proper and the transformer. The end of cord leading to headlight proper shall be capable of easy assembly and disassembly from surgical field in order to offer maximum flexibility of lamp housing and cause no interference in the surgical field. Electrical equipment, including the transformer, shall possess electrical characteristics in accordance with NFPA No. 56A requirements.

3.3.2 Lamps. Lamps shall be of the closely wound filament type matched to the fixed voltage of the transformer. Lamps shall have a rated amperage of 1 ampere, with a voltage rating of 8 volts, and a minimum life of 25 hours. Lamps shall provide a square of light in the central field of the lamp to reduce filament shadow to a minimum. Lamps exhibiting a dark, non-illuminated area in the central beam of the surgical field shall be rejected. Headlight shall be supplied with an extra lamp.

3.4 Identification marking. Each headband assembly (any of the headband assembly components) and transformer of the headlight assemblies shall be permanently and legibly marked with the manufacturer's name or registered trademark. In addition, a cautionary statement shall be permanently and prominently placed on the transformer which shall state words to the effect:

"IMPORTANT. THIS TRANSFORMER MUST BE SUSPENDED ABOVE 5 FEET FROM THE FLOOR TO BE OUTSIDE THE HAZARDOUS AREA."

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3.5 Case. Complete unit, including extra lamp, shall be furnished in a suitable case. The inside of the case shall be designed to retain components securely in place. Case shall be provided with a suitable handle, catch locks, and hinges.

3.6 Instructions for use. An instruction manual indicating instructions for use and maintenance shall be furnished with each unit.

3.7 Workmanship. Workmanship shall be first class throughout. Headlight assemblies shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the transformer and lamps may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one headlight.

TABLE I. Sampling criteria for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The headlight assemblies shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and II. Examination, however, shall not be restricted to defects listed in table II. Any headlight in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE II. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Mirror broken or cracked.
102	Headband not lined with removable, washable suspension lining.
103	Headband adjustment not self-locking.
104	Focus of headlight changes when mirror assembly is adjusted.
105	Mirror assembly not capable of vertical and lateral adjustment.
106	Mirror not capable of removal from mirror housing.
107	Headlight does not exhibit sharp illumination, free from shadows, when assembly is connected to a 110 volt, 60 Hz, AC power source.
108	Plug not in accordance with ANSI C73.11.
109	Warning tag not secured to power cord alerting personnel to remove provided plug when unit is to be used in hazardous location.

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TABLE II. Classification of defects (continued).

Categories	Defects
110	Transformer not marked with cautionary statement as specified in 3.4
111	Bayonet connector does not effect secure electrical connection.
112	Extra lamp not provided.
113	Case not furnished.
114	Power cord not free from cracks, breaks or frayed insulation.
115	External components not free from sharp edges, burrs, cracks, or rough surfaces.
116	Dimensions of power cord not in accordance with drawing.
Minor	
201	Headband and transformer not marked with manufacturer's name or registered trademark.
202	Instruction manual missing.

4.3.1 Packing inspection. The inspection of the preservation-packaging shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A, B, or C as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Cleaning. Each light shall be thoroughly cleaned in accordance with the applicable method outlined in MIL-P-116.

5.1.1.2 Preservatives. Preservatives shall not be utilized.

5.1.1.3 Unit package. Each light along with accessories shall be placed in their respective compartments within the carrying case. The carrying case shall be further packaged in accordance with MIL-P-116, method 1C-2.

5.1.2 Level C. Lights shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Six (6) lights shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicbale paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Six (6) lights shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged lights shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.2 National stock number coverage. This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-074-4190	LIGHT, HEAD, DIAGNOSTIC, Lempert-Storz, Solid Mirror, 110 Volt, 60 Hz, AC.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6515-2087

Review activity:
DOD-MB

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-L-37704 LIGHT, HEAD, DIAGNOSTIC, LEMPERT-STORZ

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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